

Danielle A. Ripsman

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 Google Scholar

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Research Summary: Large-scale and robust optimization for healthcare delivery systems; bridging the gap between inverse optimization and AI and machine learning; radiation therapy treatment planning; optimization under uncertainty (robust and Pareto-robust methods, as well as mixed integer and decision diagram-based approaches); clinical decision support and human-in-the-loop planning tools.

■ ACADEMIC APPOINTMENTS

Johns Hopkins University 2025 – Present	Malone Postdoctoral Fellow, Malone Center for Engineering in Healthcare Whiting School of Engineering, Baltimore, MD, USA Additional Affiliations: Center for Systems Science and Engineering (CSSE), Hopkins Business of Health Initiative (HBHI) Mentors: Dr. Kimia Ghobadi and Dr. Todd McNutt
University of Toronto 2024 – 2025	Postdoctoral Fellow in Operations Management & Statistics Rotman School of Management, Toronto, ON, Canada Additional Affiliations: Schulich School of Business, York University; Department of Management, University of Toronto Scarborough Mentors: Dr. Andre Cire and Dr. Adam Diamant

■ EDUCATION

University of Waterloo 2018 – 2024	Ph.D. in Management Sciences Management Science & Engineering Department, Waterloo, ON, Canada Advisor: Dr. Houra Mahmoudzadeh Thesis: Large-Scale Optimization Approaches for Radiation Therapy
University of Waterloo 2016 – 2018	M.A.Sc. in Management Sciences Management Sciences Department, Waterloo, ON, Canada Advisor: Dr. Houra Mahmoudzadeh Thesis: Robust Direct Aperture Optimization Methods for Cardiac Sparing in Left-Sided Breast Cancer Radiation Therapy
University of Toronto 2010 – 2016	B.A.Sc. in Industrial Engineering Mechanical & Industrial Engineering Department, Toronto, ON, Canada Graduated with Honours and an Engineering Business Minor

■ PUBLICATIONS

- Refereed Journal Articles**
1. **D. A. Ripsman**, S. H. Hristov, G. Gola, W. Kwong, M. Singh, E. Osei, J. Darko and H. Mahmoudzadeh. “A comparison of voxel sampling approaches for intensity-modulated radiation therapy.” *Medical Physics*, 52(12):e70179, 2025. [Link]
 2. **D. A. Ripsman**, F. Rahimi, H. Abouee-Mehrizi and H. Mahmoudzadeh. “Light Pareto robust optimization for IMRT treatment planning.” *Medical Physics*, 50(5):2637–2648, 2023. [Link]
 3. **D. A. Ripsman**, T. G. Purdie, T. C. Y. Chan and H. Mahmoudzadeh. “Robust direct aperture optimization for radiation therapy treatment planning.” *INFORMS Journal on Computing*, 34(4):2017–2038, 2022. [Link]
 4. **D. A. Ripsman**, D. M. Aleman and K. Ghobadi. “Interactive visual guidance for automated stereotactic radiosurgery treatment planning.” *Expert Systems with Applications*, 42(21):8337–8348, 2015. [Link]

Under Review 1. F. Rahimi, **D. A. Ripsman**, H. Abouee-Mehrizi and H. Mahmoudzadeh. "Superior light Pareto robust optimization with application to radiation therapy." *Manufacturing & Service Operations Management*. **Resubmitted following a major revision; under review.**

Working Papers 1. **D. A. Ripsman**, A. Cire and A. Diamant. "A decision diagram approach to multi-period home care scheduling under uncertainty." Manuscript in preparation.
2. **D. A. Ripsman**, S. A. Alev and H. Mahmoudzadeh. "A geometric approach to beam angle selection in radiation therapy treatment planning." Manuscript in preparation.
3. **D. A. Ripsman** and H. Mahmoudzadeh. "Closed-form constraint generation for classes of robust polyhedral uncertainty." Manuscript in preparation.

■ PRESENTATIONS

Conference Presentations 1. **D. A. Ripsman**, R. Chiplunkar and K. Ghobadi "The Impact of AI Scribe on Primary Care Workload: A multi-site DiD cohort study", INFORMS Healthcare, Raleigh, NC, July 2026.
2. **D. A. Ripsman** and K. Ghobadi "Generative Inverse Optimization for Healthcare Decision Making", INFORMS Healthcare, Raleigh, NC, July 2026.
3. **D. A. Ripsman**, R. Chiplunkar and K. Ghobadi "The Impact of AI Scribe on Primary Care Workload", CORS Annual Meeting, Kingston, ON, June 2026.
4. **D. A. Ripsman**, A. Cire and A. Diamant. "Multi-Period Home Care Scheduling Under Uncertainty", CORS Annual Meeting, Edmonton, AB, June 2025.
5. **D. A. Ripsman**, "A Geometrical Approach to Beam Angle Optimization", ICS Conference, Toronto, ON, March 2025.
6. **D. A. Ripsman**, F. Rahimi, H. Abouee-Mehrizi & H. Mahmoudzadeh, "Light Pareto Robust Optimization for IMRT Treatment Planning", CORS Annual Meeting, London, ON, June 2024.
7. **D. A. Ripsman**, "Integrating Uncertainty Mitigation into Radiation Therapy Treatment Planning", INFORMS Annual Meeting, Phoenix, AZ, October 2023.
8. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning", INFORMS Healthcare, Toronto, ON, July 2023.
9. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning", CORS/Optimization Days, HEC Montréal, QC, May 2023.
10. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning", INFORMS Annual Meeting, Indianapolis, IN, October 2022.
11. **D. A. Ripsman**, H. Mahmoudzadeh, "Structural Solution Approaches to Robust Optimization with Polyhedral Uncertainty", CORS/INFORMS International, Vancouver, BC, June 2022.
12. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "Closed-form Constraint Generation for Classes of Robust Polyhedral Uncertainty", INFORMS Annual Meeting, Anaheim, CA (Hybrid), October 2021.
13. **D. A. Ripsman**, S. A. Alev, H. Mahmoudzadeh, "A Geometric Approach To Beam Angle Selection In Radiation Therapy Treatment Planning", INFORMS Annual Meeting, Anaheim, CA (Hybrid), October 2021.
14. **D. A. Ripsman**, H. Mahmoudzadeh, "Closed-form constraint generation for robust radiation therapy treatment planning", CORS Annual Meeting, (Virtual), May 2021.

15. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, "Robust Direct Aperture Optimization for Radiation Therapy Treatment Planning", INFORMS Healthcare, Cambridge, MA, July 2019.
 16. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, "Robust Direct Aperture Optimization for Radiation Therapy Treatment Planning", Canadian Healthcare Optimization Workshop, Saskatoon, SK, May 2019.
 17. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, "Robust Direct Aperture Optimization for Radiation Therapy Treatment Planning", CORS Annual Conference, Halifax, NS, June 2018.
 18. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, "Robust Mixed Integer Optimization for Radiation Therapy Treatment Planning with Delivery Constraints", Optimization Days, Montreal, QC, May 2017.
 19. **D. A. Ripsman**, K. Ghobadi and D. M. Aleman. "Interactive visual decision support for radiation therapy treatment planning", Industrial & Systems Engineering Research Conference (ISERC), Montreal, QC, June 2014.
 20. **D. A. Ripsman**, K. Ghobadi and D. M. Aleman. "Interactive visual decision support for radiation therapy treatment planning", Undergraduate Engineering Research Day (UnERD), University of Toronto, ON, August 2013.
- Second place in the Industrial Engineering Oral Presentations category.**

- Competitions**
1. F. Rahimi, **D. A. Ripsman**, H. Abouee-Mehrizi, H. Mahmoudzadeh, "Superior and Light Pareto Robust Optimization", Canadian Operational Research Society (CORS) Conference Student Paper Competition, CORS Annual Conference, London, ON, June 2024.

Finalist in Health Care Operations Research SIG and Open categories.

2. **D. A. Ripsman**, H. Mahmoudzadeh, T. C. Y. Chan, T. G. Purdie, "Warm-Started Robust Direct Aperture Optimization for Cancer Treatment Planning", CORS Health Care Operational Research SIG Student Presentation Competition, CORS Annual Conference, Saskatoon, SK, May 2019.

Competition Winner.

3. University of Waterloo Three-Minute Thesis Competition, Waterloo, ON, Canada, 2018.

Selected to represent the Management Science department as finalist.

- Poster Presentations**
1. R. Chiplunkar, **D. A. Ripsman** and K. Ghobadi. "The Impact of AI Scribe on Primary Care Workload", Department of Medicine and Whiting School of Engineering Research Retreat, Baltimore, USA, 2026.
 2. R. Chiplunkar, **D. A. Ripsman** and K. Ghobadi. "The Impact of AI Scribe on Primary Care Workload", Johns Hopkins Research Symposium on Engineering in Healthcare, Baltimore, USA, 2025.
 3. **D. A. Ripsman**, K. Ghobadi and D. M. Aleman. "Graphical Interface for Interactive Parameter Selection in Radiation Therapy Treatment Planning", MIE Research Symposium, University of Toronto, Toronto, Canada, June 2013.

■ AWARDS & FELLOWSHIPS

Fellowships & Scholarships

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|-----------------------|---|
| <i>2025 – Present</i> | ■ Malone Postdoctoral Fellowship, Malone Center for Engineering in Healthcare, Johns Hopkins University |
| <i>2019 – 2022</i> | ■ NSERC Canada Graduate Scholarship–Doctoral (CGS-D) (\$105,000; \$35,000 for 3 years) |
| <i>2018 – 2022</i> | ■ Engineering Excellence PhD Fellowship (\$120,000; \$30,000 for 4 years) |
| <i>2018</i> | ■ Provost's Doctoral Entrance Award for Women (\$5,000) |
| <i>2017</i> | ■ University of Waterloo Graduate Scholarship (\$750) |
| <i>2013</i> | ■ NSERC Undergraduate Student Research Award (\$4,500) |

- 2010 – 2011 ■ John Hirschorn Memorial Scholarship (\$20,000; \$10,000 for 2 years)
 2010 ■ University of Toronto Scholars Scholarship (\$5,000)

Teaching Awards

- 2020, 2022 ■ Sandford Fleming Foundation Teaching Assistant Award, for the R laboratory of the introductory statistics course, University of Waterloo

Competitions & Honours

- 2024 ■ **Finalist** (Health Care Operations Research SIG and Open categories), CORS Student Paper Competition, London, ON — “Superior and light Pareto robust optimization”
 2019 ■ **Winner**, CORS Health Care Operational Research SIG Student Presentation Competition, Saskatoon, SK
 2018 ■ **Finalist**, Three Minute Thesis (3MT), University of Waterloo — first place in the departmental heat
 2014 ■ Third place, Simulation Competition, IIE National Student Conference, Toronto, ON
 2013 ■ Second place, Industrial Engineering Oral Presentations, Undergraduate Engineering Research Day, University of Toronto, ON
 2013 ■ Third place, Operations Research Case Competition, Industrial Engineering Competitions Day, University of Toronto, ON
 2012 ■ First place, Student Design Competition, American Society of Mechanical Engineers, University of Toronto, ON

■ TEACHING EXPERIENCE

Course Instructor **MSCI 331: Introduction to Optimization — University of Waterloo**

May – Aug 2023 Sole instructor, options section; 112 upper-year undergraduates

- **Student Course Perceptions: 4.29/5 average across the four instructor characteristics** (item range 4.04–4.55; 29 of 112 responding)
- Designed and administered online auto-graded assignments, written quizzes, midterms and final examinations; supervised two teaching assistants
- Held full administrative responsibility for the section, including announcements, grade entry and documentation

Teaching Assistant **Department of Management Science & Engineering, University of Waterloo**

Twelve terms across eight courses, 2017–2023

- 2019 – 2023 ■ *MSCI 251: Probability & Statistics for Engineers I* — designed and taught the applied R laboratories, and rebuilt them for remote delivery during the pandemic (5 terms; Dr. H. Mahmoudzadeh)
 2023 ■ *MSCI 734: Network Models* — assignment and project grading (Dr. F. Gzara)
 2022 ■ *MSCI 435: Advanced Optimization Techniques* — tutorials, grading and student support (Dr. S. Elhedhli)
 2022 ■ *MSCI 703: Applied Optimization* — grading and solution manuals (Dr. S. Elhedhli)
 2019 ■ *MSCI 401/402: Design Project* — course logistics and discussion sessions (Dr. A. Hurst)
 2018 ■ *Work Report Marking* — assessment and written feedback on work-term reports (D. Chan)
 2017 ■ *MSCI 436: Decision Support Systems* — built and taught weekly Visual Basic tutorials, office hours, grading; 42 students (Dr. M. Karimi)

- 2017 ■ *MSCI 331: Introduction to Optimization* — tutorials, office hours, grading and posted solutions; 41 students (Dr. A. Hiassat)

Courses Prepared to Teach Linear and integer programming; advanced/large-scale optimization; optimization under uncertainty (robust and stochastic); probability and statistics for engineers; simulation; healthcare operations and analytics; decision support systems.

■ RESEARCH MENTORSHIP

Undergraduate Researchers	Department of Management Science & Engineering, University of Waterloo <i>Six undergraduate co-op students and research assistants supervised, 2018–2023</i>
2023	■ Gatik Gola (co-op) — motivated the project, recommended methods and ran weekly meetings for a CT voxel sampling study; <i>co-author, Medical Physics 2025</i>
2023	■ Karina Wilk (co-op) — onboarding and weekly planning for the hospital optimization pipeline begun during the Grand River Hospital internship
2021 – 2022	■ William Kwong (research assistant) — proposed and oversaw an interface for running sampling protocols for fluence map optimization; <i>co-author, Medical Physics 2025</i>
2021	■ Tiantian Li (research assistant) — exploratory project adapting open-source radiation therapy tools (CERR, matRad) to optimization software
2019	■ Gogilan Selvarajah (research assistant) — pair programming and project direction for a sliding-window IMRT visualization
2018	■ Jessie Yeung (research assistant) — background research and data acquisition for an IMRT sliding-window modelling project

■ RESEARCH & CLINICAL EXPERIENCE

Graduate Researcher <i>Sept 2016 – Aug 2024</i>	Department of Management Science & Engineering, University of Waterloo <i>Supervised by Dr. Houra Mahmoudzadeh.</i> ■ Built optimization models for radiation therapy treatment planning, tailored to breathing motion uncertainty in breast cancer treatment ■ Prototyped and tested models against large clinical datasets using a custom C++/CPLEX code base ■ Implemented structured visualizations of plan quality in MATLAB
Clinical Research Intern <i>Oct – Dec 2018</i>	Medical Physics Department, Grand River Hospital, Kitchener, ON <i>Supervised by Dr. Ernest Osei.</i> ■ Built the I/O bridge between Varian Eclipse and open-source CERR modules for importing clinical DICOM plans ■ Wrote custom collimator rotation code and validated the pipeline on patient plans
Thesis Student <i>Sept 2015 – Apr 2016</i>	Toronto Intelligent Decision Engineering Laboratory, University of Toronto <i>Supervised by Dr. J. Christopher Beck.</i> ■ Developed A*-based batch scheduling algorithms, with C++ implementations of Dijkstra and A*
Capstone Student <i>Sept 2015 – Apr 2016</i>	Addiction Medicine Services, Centre for Addiction and Mental Health, Toronto, ON <i>Supervised by Dr. Michael Carter.</i> ■ Streamlined clinic scheduling and resource allocation, and built a simulation to validate the recommended interventions

- Undergraduate Researcher** **Medical Operations Research Laboratory, University of Toronto**
Fall 2012 – May 2014 *Supervised by Dr. Dionne Aleman.*
- Took two interactive radiosurgery planning prototypes from concept to completion in self-taught MATLAB
- Case Study Author** **Industrial Engineering Department, University of Toronto**
Fall 2014 *Case adopted as a teaching aid in Cases in Operational Research (MIE367).*
- Wrote a multi-criteria decision-making case around a restaurant venture, with realistic data distributions and a financial planning simulation

■ PROFESSIONAL SERVICE & LEADERSHIP

- Journal Refereeing** Referee for *Computers & Operations Research*, *European Journal of Operational Research*, *International Transactions in Operational Research*, *Medical Physics*, *Operations Research Forum* and *Quantitative Imaging in Medicine and Surgery*; reviewer for the Hawaii International Conference on System Sciences (HICSS)
- Conference Service**
- Session Chair, *Data-Driven Optimization in Healthcare*, INFORMS Computing Society (ICS) Conference, Toronto, ON, 2025; also a conference volunteer
 - Session Chair, CORS/INFORMS International Conference, Vancouver, BC, 2022
 - Designed the virtual conference venue (Gather Town) for the CORS Annual Conference, 2021, with Dr. Kiefer Burgess
- Memberships** INFORMS (Health Applications Society); Canadian Operational Research Society (CORS)
- National Student Councillor** **Canadian Operational Research Society (CORS)**
2024 – 2025 *Elected to represent the national student body.*
- Led the post-pandemic revival of student chapters across Canada
- President** **The Operations Research Challenge (TORCH), University of Waterloo**
Sept 2017 – 2020 *Founded the Waterloo-region edition of the annual high school competition.*
- Led logistics for the 2018, 2019 and 2020* competitions: student and volunteer recruitment, funding, testing and scheduling
 - Wrote competition problems and validators, and coordinated the automated grading software, with volunteers from Waterloo, Concordia and Toronto
- Student Chapter President** **Canadian Operational Research Society (CORS), Waterloo, ON**
Sept 2017 – Sept 2018 *Ran the local student chapter's speaker and social programme.*
- Coordinated speakers and events, onboarding, communications and website oversight
 - Grew active membership from fewer than 10 to nearly 30
- President** **Management Sciences Student Association, University of Waterloo**
Sept 2016 – Aug 2017 *Elected departmental representative for graduate students.*
- Budgeted and ran invited speaker and social events; represented students at monthly departmental and cross-departmental meetings
- Executive Member** **The Operations Research Challenge (TORCH), University of Toronto**
Fall 2014 – Winter 2016 *Co-Chief Creative Officer (2016); Assistant Editor-in-Chief (2015).*
- Oversaw creation of the 2016 contest, edited the 2015 problem set, and wrote a scheduling problem with an automated solver for grading

*Cancelled due to the pandemic

■ INDUSTRY EXPERIENCE

Rational Software Developer <i>May 2014 – Aug 2015</i>	IBM Canada Software Lab, Markham, ON <i>Eclipse-based enterprise software development, testing and debugging (TPF Toolkit, RDi); supervised by Rob Cecco.</i>
Engineering Intern <i>Summer 2012</i>	Rotork Controls Canada, Mississauga, ON <i>Engineering drawings, customer documentation, and a Visual Basic database for intracompany drawing retrieval.</i>

■ TECHNICAL SKILLS

Languages & Tools	Python, C++, MATLAB, R, $\LaTeX$
Optimization	CPLEX, Gurobi, AMPL
Clinical / Domain	Varian Eclipse, CERR, matRad, DICOM-RT

Last updated: October 7, 2026